

Marcelo da Fonseca Oliveira

Continuous Assessment:

Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines

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Assignment submitted in partial fulfillment of the requirements for the Applied Machine Learning in course in the Data Analytics program at City College Dublin – Dublin Campus.

Professor: Manuel Benitez

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Continuous Assessment: Flu Shot Learning: Predict H1N1 and Seasonal Flu Vaccines

Course: Applied Machine Learning

Professor: Manuel Benitez

Student Name: Marcelo da Fonseca Oliveira

Student ID: 25037510

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Declaration:

I hereby declare that this assignment is my own work and has not been submitted, either in whole or in part, for any other academic purpose. All sources of information have been appropriately acknowledged.

Signature: Marcelo da Fonseca Oliveira

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1. Business Context

Influenza vaccination has been a public health challenge due to low coverage rates, despite vaccine availability. Dong et al. (2025) noted that China vaccinated 6.75% of its adults, a worrying number compared to developed countries. This scenario highlighted the need for predictive models that identify individuals with a low probability of vaccination.

According Wang et al. (2024), Logistic regression proved adequate when comparing seven machine learning algorithms to predict childhood vaccination. The results showed that this method is more suitable due to better interpretability for clinical health professionals, even though Random Forest performed better on the training set. Logistic regression used variables that minimized the indicators, Akaike Information Criterion (AIC) the metrics Accuracy, Cohen's k , and Area Under the Curve (AUC).

However, Wang et al., (2024), also expo that Random Forest presented optimized hyperparameters ($n_{tree}=180$, $max_depth=15$, $min_samples_split=10$), which allows identifying a characteristic through %Inc. This contrasts with logistic regression and its lower clinical interpretability, the difference in the transparency of decisions.

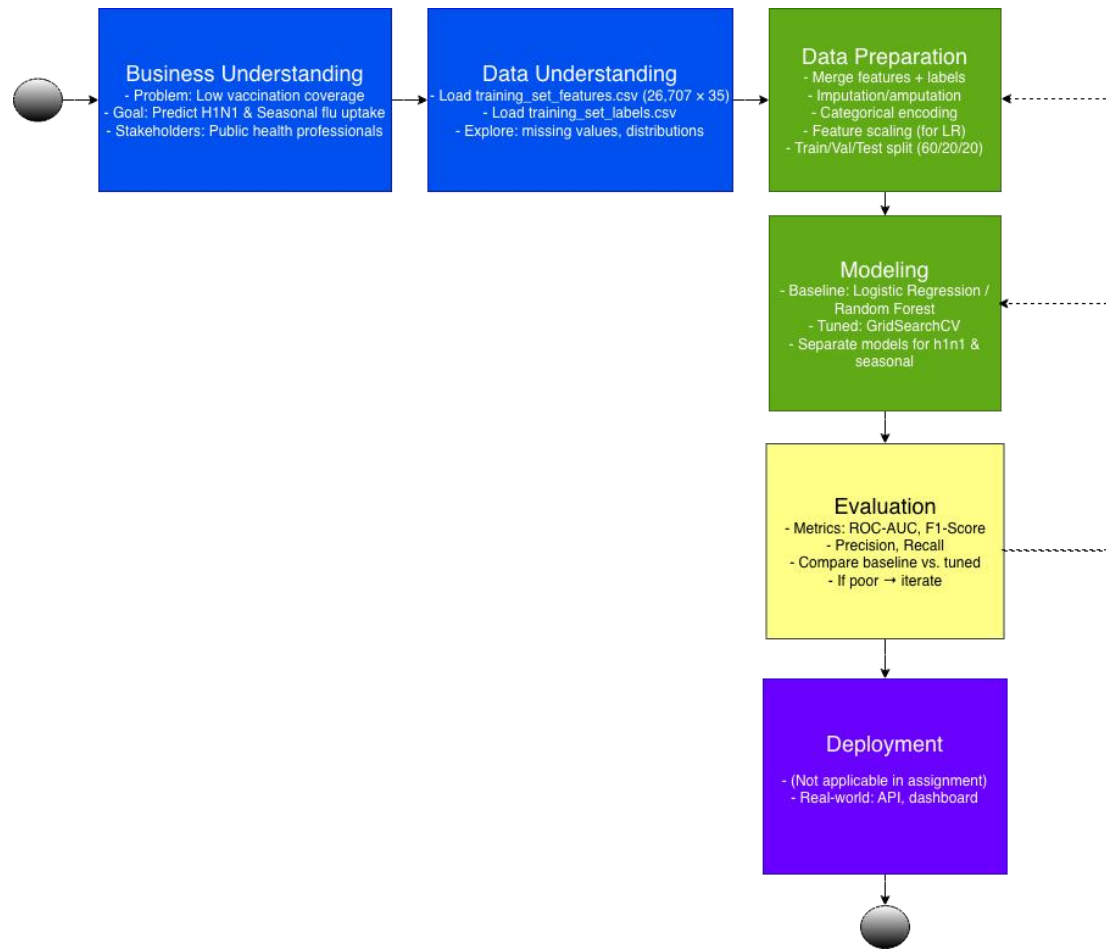
Choi et al. (2026) also corroborate logistic regression, due to the effectiveness it showed in a vaccination system ($VE=20.4\%$, 95% CI: 8.2-30.9), the study presented, in a test-negative design (TND), that linear methods still present a better choice in epidemiological studies.

2. Methodology Framework

Thinking about a public health context, for this project It was chosen the CRISP-DM (Cross-Industry Standard Process for Data Mining) methodology according Plotnikova (2021), which has an iterative and guided approach that has six phases, as show in the flowchart below: Business Understanding (defining low vaccination coverage as the core problem), Data understanding (exploring the National 2009), Data Preparation (handling missing values, categorical encoding, and stratified train/validation/test splitting), Modeling (implementing baseline and GridSearchCV-tuned Logistic Regression and Random Forest), (assessing ROC-AUC, F1-score, precision, and recall), and Deployment (not applicable in this academic context).

For Ploeg (2023), owed to the iterative refinement between the modeling and evaluation phases, this structure presenting to be the most suitable for forecasting problems. Due to a complexity of 30 variables with missing values, data preparation proved to be more challenging. Data integrity was maintained, which required more cautious decisions when it came to the imputation strategy, in order to avoid incorporating biases that could compromise the validity and reliability of the model.

Figure 1: Logistic Regression: Flowchart explain the process



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